PPL FINAL

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① Consider the following C declaration, compiled on a 32-bit Pentium machine:

```
struct E
int n;
char c;
} A[10][10];
```

If the address of A[0][0] is 1000(decimal), what is address of A[3][7]?

Answer: $1000 + (3 \times 10 \times 8) + (7 \times 8) = \boxed{1296}$

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②
```
int sub(int *i){
  *i+=2;
return(*i)+3; }
void main(){
i=1,j=1,s1,s2;
s1=sub(&i)+(i+1);
s2=(j*5)+sub(&j)
```

a) Left to right
b) Right to left

---

③ Consider where STR is token

$A \rightarrow A \ \Delta \ B \mid B$

$B \rightarrow C \ \square \ B \mid C$

$C \rightarrow C \ \Diamond \ D \mid D$

$D \rightarrow STR \ \heartsuit \ D \mid STR$

Associality of operators?

Δ: Left
$\square$: right
$\Diamond$: left
$\heartsuit$: right

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(4)

```
Procedure main(){
  string x;
Procedure print_message(){
  Print(x); 3
procedure sub1{
  string x="sub1_message"
  sub2(print_message); 3
Procedure sub2(sx:procedure){
string x="sub2 message";
 sx(); 3
x="main message";
 sub1()
```

a) shallow Binding

Ans: sub2 message

b) Deep Binding

Ans: Main message

(5)

```
begin
integer n;
procedure P(k:integer);
 begin
n:=n+1;
k:=k+4;
print(n);
 end;
```

a) Call by value: 1 1

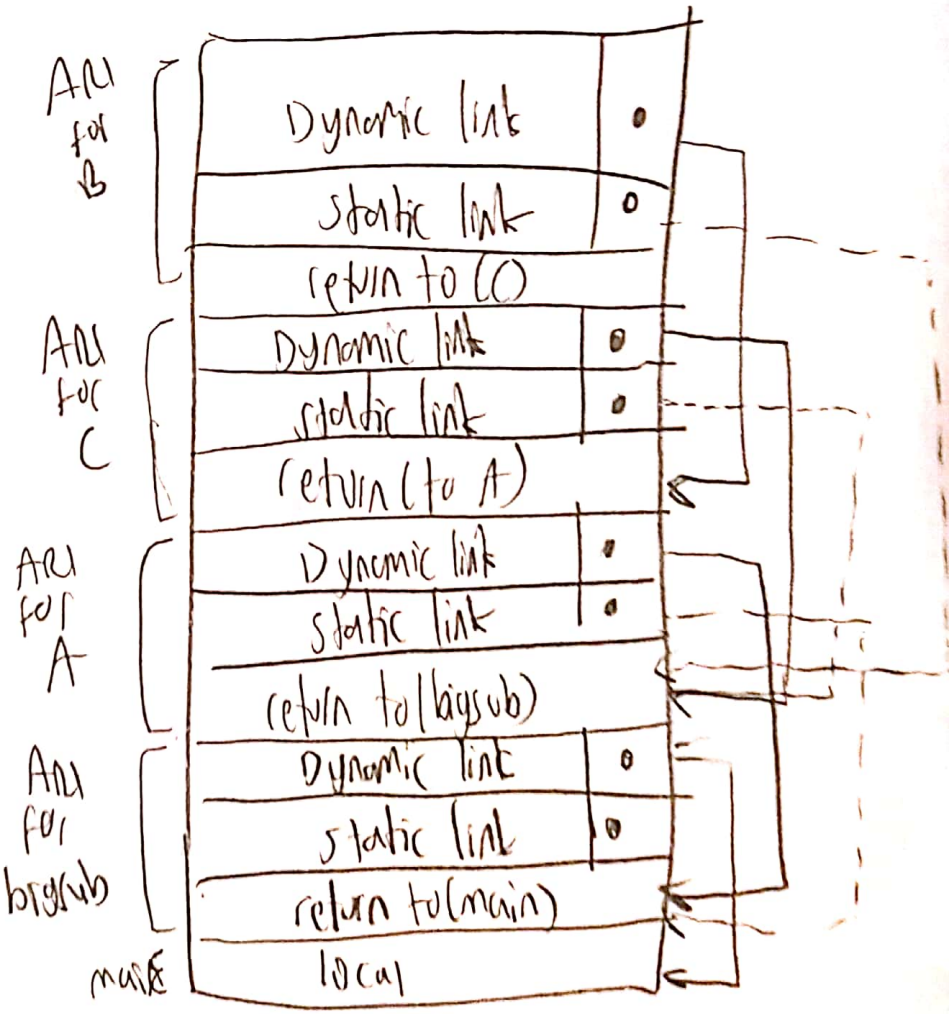
b) Call by reference: 5 5

c) Call by value-result: 1 4

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(6)

```
procedure Main is
procedure Bigsub is
Procedure A is
Procedure B is
 begin --- of B
  ---  <-------- 1
  end; -- of B
 procedure C is
 begin; -- of C
  --
  B;
 --
 end; -- of C
 begin --of A
  --
  C
 --
 end; -- of A
begin -- of Bigsub
 --
  A;
 --
end; -- of bigsub
```



Answer:

call sequence

$main calls bigsub
bigsub calls A
A calls C
C calls B

Dynamic link → call sequence direction.

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